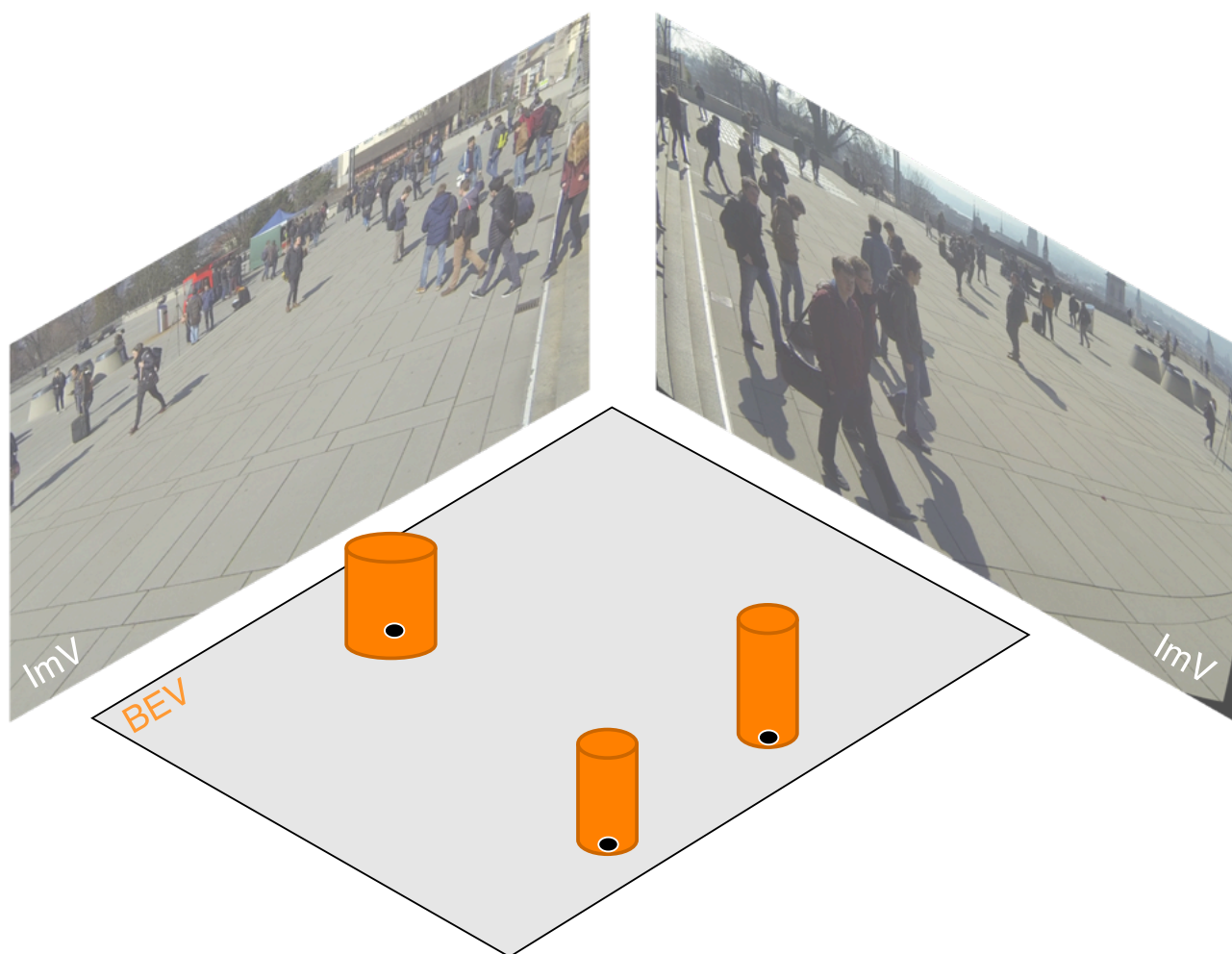


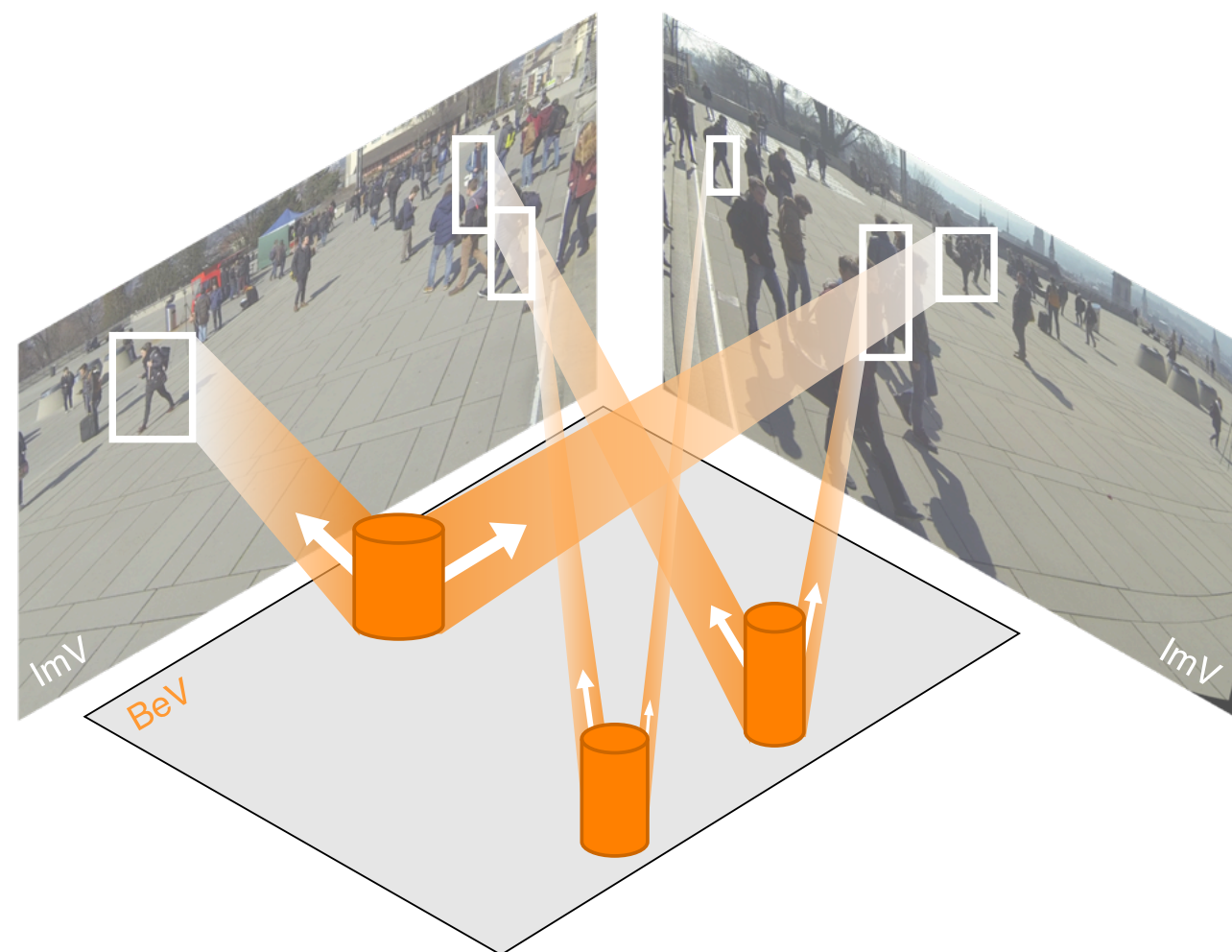
Step 1: Cylindrical Query Initialization

Initialize a set of 3D cylindrical queries across the BEV. Each query encodes an object hypothesis.



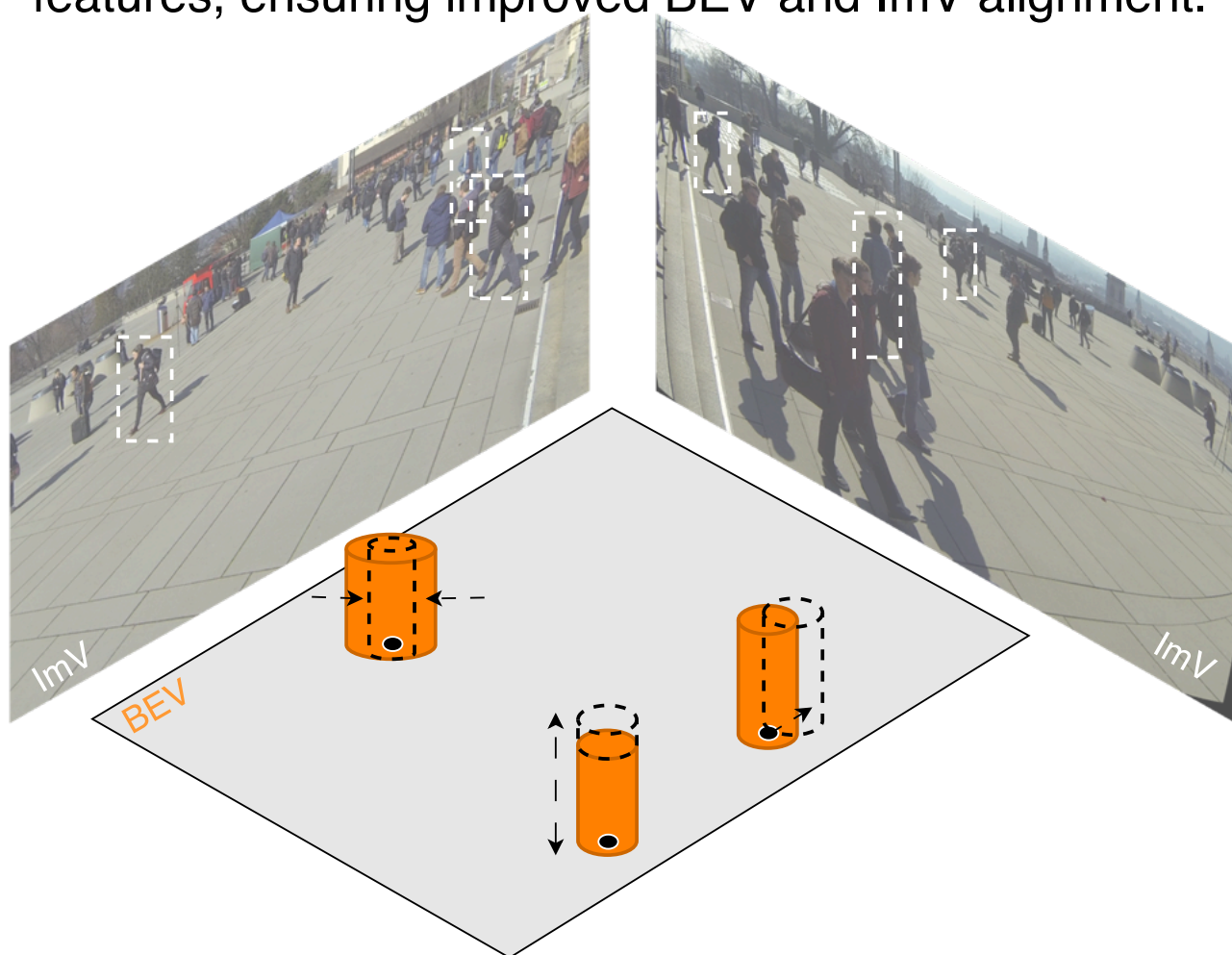
Step 2: Feature Sampling & Aggregation

Project each 3D cylindrical query into all camera views. Aggregate multi-view image features.

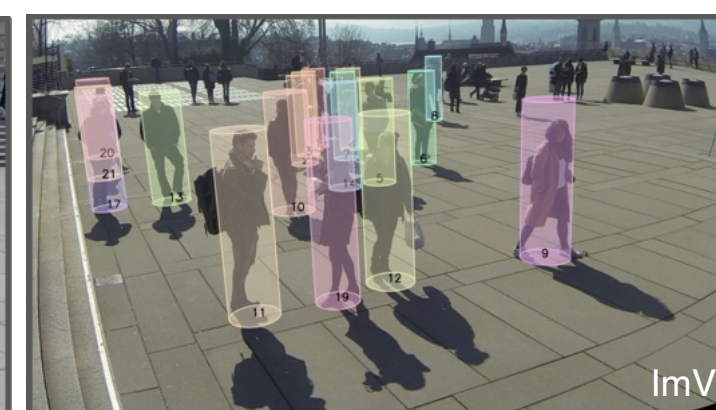
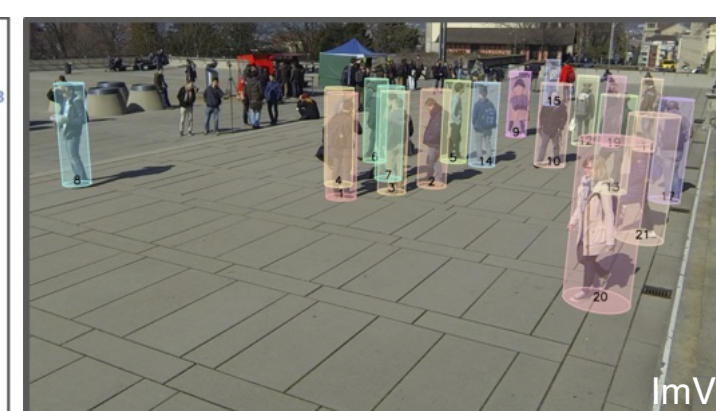


Step 3: Cylinder Position & Shape Refinement

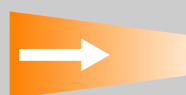
Refine cylindrical query using the aggregated multi-view features, ensuring improved BEV and ImV alignment.



MVCyDe: 3D-BeV & 2D Image-view Detection & Tracking



BEV: Bird's-Eye-View ImV: Image-View



Projection Direction

• Absolute Ground Point



3D Cylindrical Query



Projected bounding box from Cylindrical Query